ICTCE04

International College
Rangsit University
52/347 Muang-Ake, Phaholyothin Road, Lak-Hok, Muang, Pathumthani 12000, Thailand
T. (662) 997-2200-30 Ext. 4012-4020, F. (662) 997-2200-30 Ext. 4016
www.rsuip.org

Cooperative Education Liaison:
Miss Srinuan Ketkaew
T. (662) 997-2222 Ext. 4012 E-mail: srinuan@rsu.ac.th

Weekly Progress Report Form

Name - Mr. Sai Swam Wan Hline, Student ID - 6609216

Tel. 0649435041, E-mail - saiswamwan.h66@rsu.ac.th,

Company Name /Employer Name - **HTZKKL Technologies Co.,Ltd.**

Reporting Period: *Week No.4, September 2026, 23/9/2026 - 29/9/2026, Total 40 Hours*

1. Name of project/task/issue/topic of student's responsibility.

(List each of the performed tasks.)

Project: Multi-Tenant School Intranet & Academic Management Platform

Task 1: Full-stack design and integration of the Bilingual Internationalization (i18n) Engine supporting English and Myanmar Unicode.

Task 2: Database schema localization with Alembic migrations (a1b2c3d4e5f6) adding Subject.name_mm and User.language_preference.

Task 3: Architecture of 3-Way Official Document Mode Switchers (Official Myanmar, English, and Side-by-Side Bilingual) for official A4 printable dossiers and receipts.

Task 4: Bilingual message template composers for automated Telegram and Viber parent notification bots.

Task 5: Modernization of the entire frontend iconography using the lucide-react vector SVG system and bilingual test suite development (test_i18n_models_and_services.py).

2. Summary of the performed tasks.

(Describe the type of work that the student has accomplished.)

During the fourth week of the project:

1.Full-Stack i18n Architecture: Designed and implemented a lightweight, typed React translation context (i18n-context.tsx) and custom hook (useTranslation()). Structured 18 domain-specific JSON translation dictionaries for both English (en) and Myanmar (my), covering all 27 dashboard routes with 100% key parity. Integrated Google Font Noto Sans Myanmar with customized typography metrics.

2.Database & User Preference Persistence: Executed an Alembic migration adding name_mm to the Subject entity for native curriculum display and language_preference to the User model, persisting user language choices across devices and browser sessions.

3.Tri-Mode Printable Dossiers & Receipts: Engineered a tri-mode document rendering toggle (Myanmar, English, Bilingual) for official printable A4 documents, including Student Registration Cards, Official Payment Receipts, and Staff Record Cards.

4.Bilingual Bot Notification Engine: Refactored Telegram and Viber notification dispatchers to generate bilingual alerts. Parents now receive exam results, daily attendance notices, and fee balance reminders formatted in their preferred language.

5.UI Vector Modernization & Icon Standards: Replaced informal emojis across 69 frontend views, tables, and modal dialogs with scalable lucide-react vector icons, enforcing strict design token guidelines while preserving country flags in the language selector.

6.Testing & QA Verification: Authored tests/test_i18n_models_and_services.py with 109/109 tests passing (100% pass rate) and validated that all 27 Next.js production routes compile cleanly with zero TypeScript errors.

3. Application of concepts, theories, knowledge, or experiences that applied to cooperative education.

Software Internationalization (i18n) & Localization (L10n): Applied modern localization engineering principles, managing modular resource bundles, dynamic pluralization, and fallback locales without hardcoding UI copy.

Typography & Complex Text Layout (CTL): Applied web typography standards for complex non-Latin scripts (Myanmar Unicode), adjusting line-heights, vertical alignment, and font-fallback chains to avoid visual clipping.

State Management & Context API: Utilized React Context and custom hooks to propagate language state changes across deeply nested components without

unnecessary re-renders.

Cross-Platform Communication Protocols: Designed localized notification pipelines adhering to Telegram Bot API and Viber REST API standards.

Scalable Vector Graphics (SVG) & Visual Design: Implemented design system icon tokens using lucide-react to ensure sharp rendering across high-DPI displays and print stylesheets.

4. Problems/obstacles encountered during the work.

(Provide details on how to solve the problems/obstacles.)

Problem 1: Text Clipping & Line-Height Collisions with Myanmar Unicode:

Detail: Due to stacked diacritics and tall consonants in the Myanmar script, default Tailwind line-heights (leading-none or leading-tight) caused diacritical marks to be clipped in table cells and navigation badges.

Solution: Adjusted typography utilities to leading-normal and leading-relaxed on localized text containers and configured custom font metrics for font-myanmar in tailwind.config.ts.

Problem 2: Layout Bleed in Printable A4 Documents When Switching Languages:

Detail: Certain Myanmar phrases are longer than their English equivalents, causing single-page official payment receipts and dossiers to push content onto an unwanted second printed page.

Solution: Implemented @media print CSS rules with compact padding, dynamic font-size clamp rules, and two-column side-by-side layouts for the bilingual mode to guarantee that official records strictly fit within one standard A4 sheet.



(……Nay Min.....)

Job Supervisor

Date ……29/9/2026.....



(….Sai Swam Wan Hline.....)

Student

Date ……29/9/2026.....